HAYDEN VAN STEYN

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EDUCATION

Lafayette College | B.S. Mechanical Engineering, Cum Laude | GPA 3.66 | May 2025
AIB, Bonn, Germany | Engineering Study Abroad: Architecture & Built Environment | Spring 2023

TECHNICAL SKILLS

CAD & Design: SolidWorks, Fusion 360, AutoCAD | 3D modeling, assemblies, 2D detail drawings, DFM/DFA
Engineering Analysis: ANSYS FEA (structural, thermal, vibration), CFD | hand calculations, failure analysis, root cause investigation, tolerance stack-up
Testing & Validation: Test protocol development and execution, iterative physical validation, test fixture design | force/sensor data acquisition, formal test reporting
Fabrication & Prototyping: CNC machining, 3D printing, sheet metal, hands-on assembly | experienced machine shop operator | rapid prototyping and design iteration
Documentation: BOM preparation, engineering drawings, technical specifications, design documentation | cross-functional team communication
Other: MS Office, Adobe CC | 3D scanning | visual arts, sketching | US citizen, authorized to work without sponsorship

ENGINEERING PROJECTS

Autonomous Ocean Search & Rescue Drone: Senior Design Project | Lafayette College | 2024–2025

- Led full product development cycle from concept specification through Fusion 360 CAD, FEA simulation, prototype fabrication, and iterative physical validation, achieving 35% weight reduction while maintaining full structural load capacity under worst-case dynamic conditions.
- Owned mechanical design of a precision pneumatic actuation mechanism with sub-5-second performance requirement; resolved structural vibration failure mode through root cause analysis and joint geometry redesign.
- Developed and executed test protocols; produced detailed BOM, 2D drawings, and milestone documentation throughout full development cycle; coordinated across mechanical, electrical, and software disciplines.

Terrain-Adaptive Race Car: Chassis, Drivetrain & Braking Design | Lafayette College | Spring 2025

- Directed SolidWorks structural design of chassis, drivetrain, and braking system under complex dynamic loading; applied DFM, tolerance analysis, and material selection across multi-component 3D assemblies with stringent weight and packaging constraints.
- Coordinated multi-discipline subsystem integration; produced detailed design documentation for cross-functional review.

Wind Tunnel Analysis: Aerodynamic Performance Study | Lafayette College | Spring 2025

- Designed and executed a structured multi-speed test program using force/sensor data acquisition; applied CFD simulation to model pressure distribution and airflow behavior; produced formal engineering report with quantitative findings and design recommendations.

FEA: Bolt Pattern Optimization | Lafayette College | 2024

- Optimized pressure vessel bolt pattern using ANSYS FEA and hand calculations to minimize bolt count while ensuring seal integrity under internal pressure loading.

WORK EXPERIENCE

Mechanical Engineering Intern | PPL Electric Utilities | PA | Summer 2024

- Collaborated with engineering teams to reorganize CAD model database and standardize cross-functional documentation workflows in a regulated technical environment; developed understanding of structured engineering processes and documentation standards.

Machine Shop Lab Assistant | Lafayette College | Easton, PA | Fall 2023

- Operated CNC machines, sheet metal equipment, and full fabrication toolset; mentored engineering students on prototyping strategies, DFM thinking, precision measurement, and safe shop practices.

Research Assistant & Community Educator | Blue Endeavors | CA | 2019–2021

- Conducted structured field research including behavioral surveys and environmental data collection; contributed to CO2 management and restoration research; led educational presentations communicating scientific findings to diverse audiences.

ACTIVITIES & INTERESTS

Certifications: CPR (Red Cross) | PADI Scuba: Open Water, Deep, Wreck, Night, Nitrox, Coral Reef Conservation | NOLS Mountaineering
Leadership: President, Lafayette Outdoor Society: led adventure programming and environmental stewardship initiatives for campus outdoor community
Engineers Without Borders | Visual Arts, Sketching & Woodworking
Outdoor pursuits: Backpacking (multi-month expeditions), skiing/backcountry, surfing, mountain biking